

SOLAR POWER SYSTEM INSTALLATION

Client 2 - Vietnam



System Design & Sourcing

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EXECUTIVE SUMMARY

The Client 2 and Solar

Due to the increased industrial development in Vietnam, wildlife has been negatively impacted in the country. As the elephant population is among the most adversely affected, the local government has initiated wildlife protection programs including the construction of animal conservation centers. The Client 2 Elephant Conservation Center is one such development currently under construction and has already begun caretaking efforts. Two rescued elephants from the local area, and one grown bull are among the first to join.

Due to an unstable energy grid supply during the months of November through April inclusive, the Client 2 wishes to install a solar power system in order to reduce demand on the local energy grid, create a back-up power source for the facility and promote sustainable energy and environmental awareness in the region. The system must be expandable to support future growth of the Client 2 organization and future power demands.

Ninety-five percent of project funding was provided as a donation by the _____ through the efforts of Client 1, PhD, director of the Client 1 Foundation of Vietnam after being approached by the Client 2 for collaboration.

Objectives

Objective 1: De-stress the local energy grid, particularly during peak use hours.
Objective 2: Design a back-up power solution should grid energy supply fail.
Objective 3: Create an income stream and reduce electricity expense for the Client 2 with the solar power system.
Objective 4: Design a solar power system that is an impressive public relations piece.

Solution

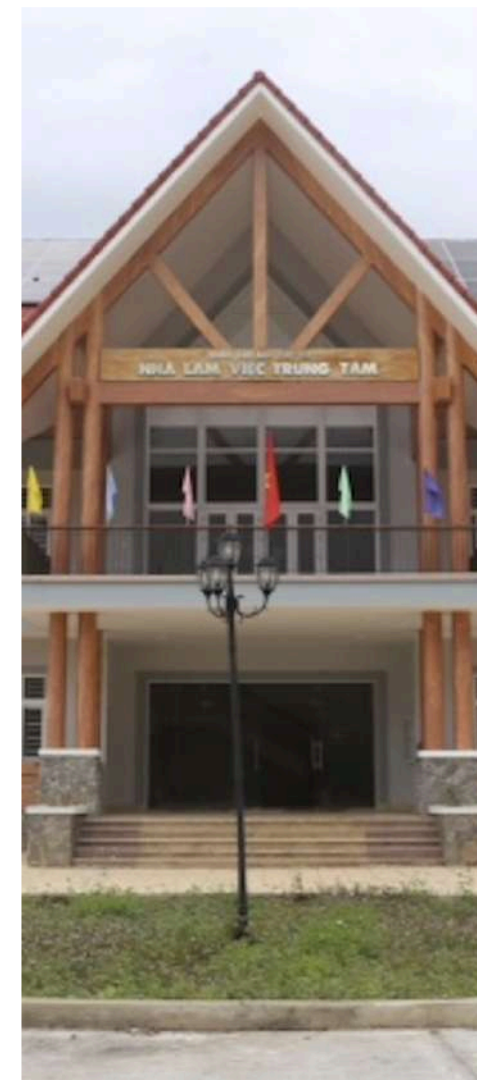
As the system requirements include the ability to sell energy to the power grid as well as provide power in case the power grid is disabled, a 41kW expandable grid tied solar power system with back-up storage using a DC-coupled solar array was selected. The power electronics control, hardware and architecture are provided by Schneider Electric's (SU.PA) solar division, a French multinational who is a specialist in energy management and automation digital solutions. The system includes two clusters of three AD/DC solar charger inverters each. The system's sixty four photovoltaic panels which make up the 21kW solar array were sourced from JA Solar, a Chinese solar company headquartered in Beijing, China with a substantial global market presence, known for high quality solar solutions. The solar array is paired with as set of

four MPPT charge controllers also from Schneider Electric. The energy storage solution consists of 32kWh of available energy from two separate sealed lead acid (SLA) battery banks provided by Vision, a Chinese company headquartered in Shenzhen, China with over 20 years experience in the energy storage space.

Results

The solar power system successfully reduces daily power demand on the grid by 29.2% (113kWh), reducing the daily average draw from 387kWh to 274kWh. This was accomplished by allowing the solar array to support the majority of peak loading in the morning and the battery bank to support peak loading in the afternoon while charging only during off-peak and standard hours. Total annual energy cost reduction is approximately US\$ 6,913, with the addition of US\$ 270 income from solar energy sold to the grid.

Considering energy savings and income stream, after accounting for equipment depreciation and operational costs, average total annual cash flow comes to US\$ 3,020. The Schneider Electric solar power system is set up in the main lobby of the Client 2 building on the first floor with a large monitor display showing live solar harvesting activity. The system was commissioned on December 23rd, 2020.



INTRODUCTION

The Client 2: Energy Needs

Overview of Client 2 Client 2 (Client 2)

The development of Client 2 started in 2011 with funding from central and Client 2 government. Since 2015, Client 1 has supported Client 2's development by providing international consultants to assist in the design and development of Client 2, and training of staff on elephant welfare and care. Once complete, the centre will be composed of two large elephant care areas. Each area consists of a round elephant house of over 1000m² and opens into enclosures surrounded by 2.5km of semi-permanent fencing. The center also contains administration and staff facility, veterinary facility, education/visitor facility, accommodation, storage and food preparation facilities and utilities to service the center. The unique design of the center allows captive elephants to transition from intensive management to extensive management and where possible back to the wild. Once fully operational, Client 2 will conduct elephant friendly tours to educate visitors on elephant protection and conservation.



Statement of Needs

The Asian elephants have a significant role in Vietnam's cultural and ceremonial activities especially among the ethnic minority groups in Vietnam's central highlands. The majority of wild and captive elephants are found in Client 2 province in Vietnam's central highlands. Rapid economic developments have contributed to major loss of the elephants natural habitat, and with ongoing illegal ivory hunting, Vietnam's wild elephant population has declined from 2000 in 1990 to less than 150 in 2019.

To arrest the extinction of elephants, the government of Vietnam has enacted the National Elephant Action Plan to protect wild elephants and to increase the captive elephant herd. The majority of captive elephants are owned by the ethnic minorities in Client 2 Province.

A major element of this plan is the establishment of an Client 2 on 200 hectares of land in Buon Don District, Client 2 Province. The aims of the centre are to provide sanctuary for wild rescued elephants, to improve the welfare and care of captive elephants, and to raise public awareness on elephant protection and conservation.

The Need for Solar Power at Client 2

Vietnam's central highlands climate is characterized by two distinct dry and wet seasons. At present most of Client 2 provincial electricity requirement is provided by 24 small scale hydro power stations. However, during the dry seasons of November to April, power shortage occurs due to over demand. Power shortage is affecting Client 2's operations of maintaining E-fence integrity, treatments of elephants, and maintaining the irrigation system for water hectares of land to grow food for elephants.

Vietnam Environment Administration agency rated four Client 2's districts of Buon Don, Ea Sup, Krong Bong, and Ea H'leo of Client 2 most suitable for solar power development. The daily solar energy for Buon Don district where Client 2 is situated is rated at just over 5.0 kWh/m²/day. Based on the above rating, solar power energy for Client 2 is highly feasible to provide Client 2's energy needs. By using solar power, the Client 2 is contributing to environmental protection through the use of a renewable energy source to run its centre, reducing pressure on local electricity demand, and helping to promote the use of clean and sustainable energy.

At present the Client 2 is caring for two wild rescued elephants, provides welfare and care support to 45 captive elephants in surrounding areas. The Client 2 has 20 staff and is using just

over 130,000kWh/year, based on the average monthly energy bills. Once the Client 2 is fully developed, it will have around 50 staff and will care for up to 20 rescued elephants at any time.

The amount of energy needed to maintain Client 2's operation includes the running of veterinary facility for medical treatment of elephants, education/visitor area, storage and food preparation area for elephants, staff accommodation, administration and staff facility, utilities to service in the centre, and irrigation system for 20 hectares of land to grow food for elephants. Although there are two large elephant care areas with enclosures surrounded by 2.5km of semi-permanent fencing and electrical fencing, the actual amount of power consumed by electrical fencing is less than 500W.



INTRODUCTION

The Client 2: Site Characteristics

Client 2 Site Map



Site Coordinates: Latitude: 12.927814, Longitude: 107.809193

1. Administration/Visitor/Services
2. Elephant Keeping Areas
3. Corridor Containing Services
4. Elephant Enclosures

Note: Distance between Location 1 and Location 3 is approximately 1200m

Client 2 Administration Building



The Client 2 Administration/Visitor/Services building is intended to house the solar power system.

- Power Management and Monitoring components will be installed near the building's main electrical panel. Area must be dry and free of excessive dust or debris.
- Battery Bank is intended to be installed near the Power Management and Monitoring components in a well ventilated and dry area.
- Solar Panel Array can be installed on the roof of the administration building, though for optimal energy harvesting should be mounted facing south at a 78° angle relative to vertical with a minimum of surrounding sunlight obstructions such as trees or other buildings.

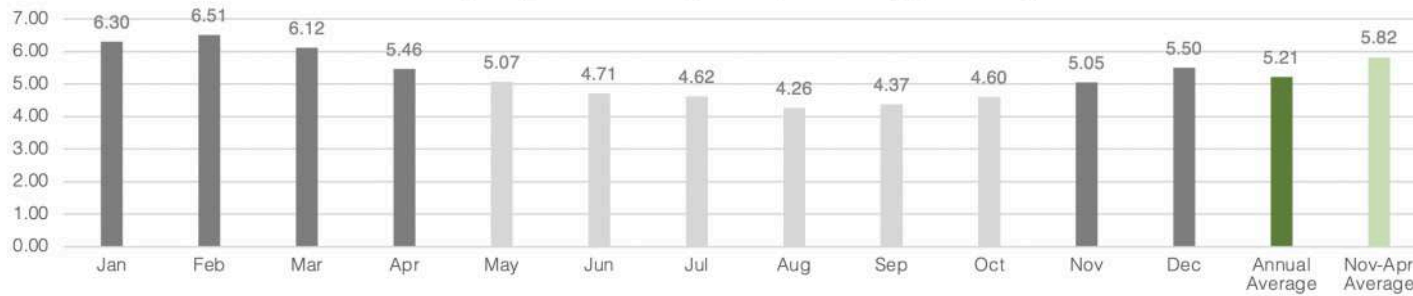
INTRODUCTION

The Client 2: Site Characteristics

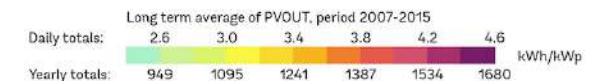
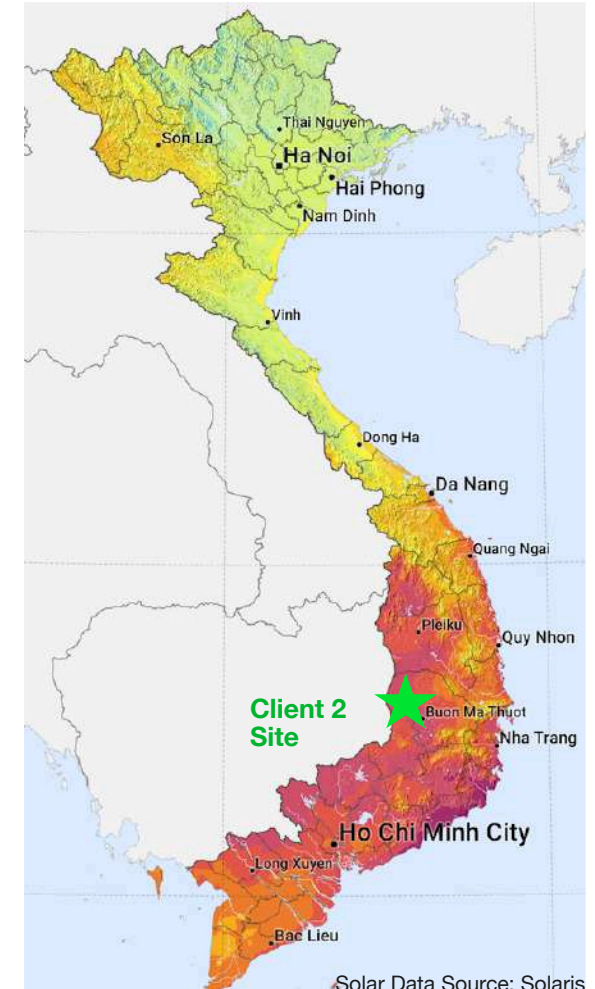
Solar Isolation

The Client 2 site experiences an annual daily average solar isolation of 5.03 kWh/m²/day. By ensuring that the solar panel array is south facing at the optimal angel for year-round solar harvesting, the annual average isolation increases 3.6% to 5.21 kWh/m²/day, and between the months of November through April, increases a total of 15.7% to 5.82kWh/m²/day.

Average Daily Solar Isolation (kWh/m²/day)



Note: Solar Isolation is the measured solar irradiation over a given period of time. In the chart above, the period is “per day”.



INTRODUCTION

The Client 2: Site Characteristics

Load Profile

The actual average daily power use requirements of the facility are confirmed with the Client 2 per the demand table on the right. Intended loads and approximate scheduling reveals an estimated daily load profile and the following key data points:

- Peak Demand: 29.7kW
- Minimum Demand: 8.7kW
- Average Demand: 16.1kW

Assumptions were made where specific scheduling information was missing. The load profile shows that the majority of loading occurs between the hours of 5:00am and 4:30pm by the pump system and water heaters. Peak demand occurs during the hour of 10:00am to 11:00am due to the pump system, air conditioners and office equipment operating simultaneously.

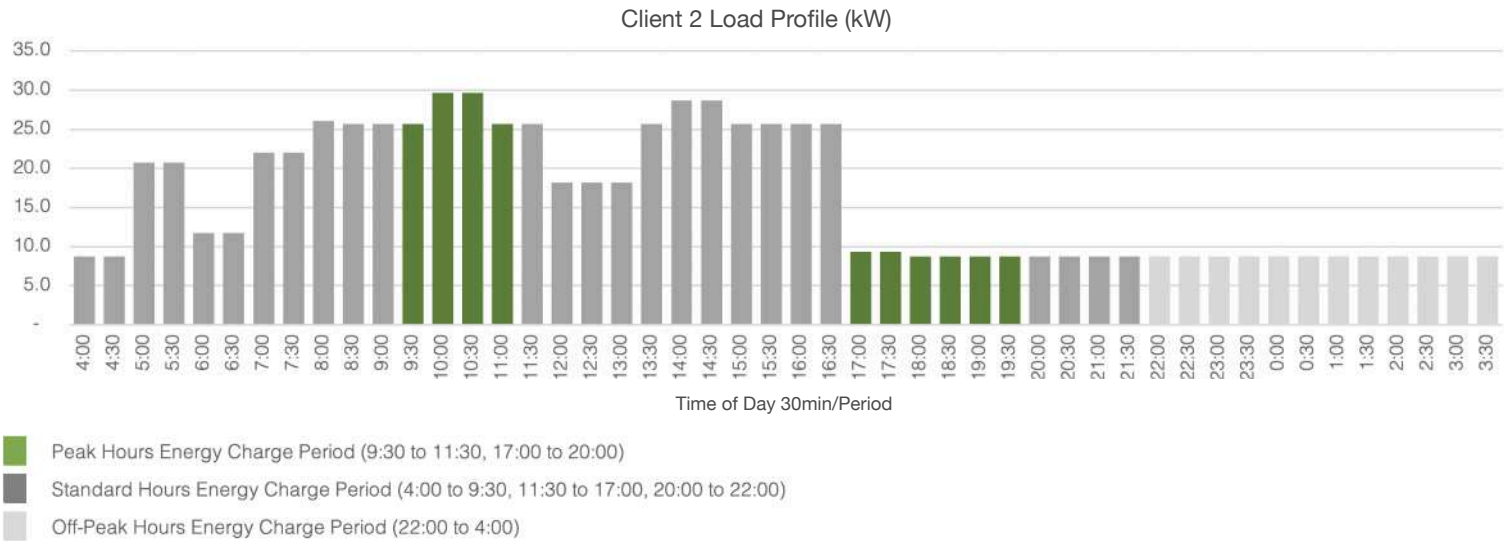
Minimum demand occurs during the hours of 6:00pm and 4:30am only. An average minimum demand could be considered at 9.9kW between the hours of 6:00pm and 7:00am.

A complete facility energy assessment is recommended in order to facilitate load shifting and assist to identify alternative low energy draw equipment, such as light bulbs, water pumps and air compressors to minimize energy demand.

Main EEC Daily Energy Demand - tổng hợp các thiết bị sử dụng điện và điện năng tiêu thụ trong ngày

STT	Device - Thiết Bị	Location	Quantity (Số lượng)	Voltage (Hiệu điện thế) (V)	Power (Công suất) (Watts)	Total Power (Tổng Công suất) (Watts)	Hours Per Day (Giờ sử dụng trong ngày)	Day/ Night (Ngày/Đêm)	Watt- hours energy (Wh)
1	E-Fence (Rào điện)	1	1	220AC	500	500	24	Both	12,000
2	Pump System (Hệ thống bơm)	1	1	220AC	7500	7500	8.5	Day	63,750
3	Pump System (Hệ thống bơm)	1	2	220AC	2000	4000	1	Morning	4,000
4	Pump System (Hệ thống bơm)	1	1	220AC	3000	3000	1	Afternoon	3,000
5	Lighting Type 1 (Bóng đèn loại 1) 3x18W	1	30	220AC	54	1620	10	Day	16,200
6	Lighting Type 2 (Bóng đèn loại 2) 3x24W	1	34	220AC	72	2448	10	Day	24,480
7	Lighting Type 3 (Bóng đèn loại 3) 1x36W	1	2	220AC	36	72	14	Night	1,008
8	Lighting Type 4 (Bóng đèn loại 3) 13W	1	28	220AC	13	364	14	Night	5,096
9	Water heaters (Máy nóng lạnh) 1500W	1	8	220AC	1500	12000	1	Various	12,000
10	Outdoor Lighting Systems (Hệ thống chiếu sáng ngoài nhà)	1	12	220AC	250	3000	12	Night	36,000
11	-80 C Freezer (Tủ lạnh âm sâu 80 độ)	1	1	220AC	2500	2500	24	Both	60,000
12	Office Equipment: Computers, Desk Lighting, Fans, Printers, etc.	1	5	220AC	560	2800	10	Day	28,000
13	Air Conditioner 1 - Office Spaces	1	4	220AC	1500	6000	10	Both	60,000
14	Air Conditioner 2 - Conference Rooms	1	2	220AC	3000	6000	1	Both	6,000
15	CCTV	1	1	220AC	1500	1500	24	Both	36,000
16	Phantom Loads	1	1	220AC	799.56	799.56	24	Both	19,189

Total/Day - Tổng (kWh):	386.72
Total Annual (kWh):	141,154.06



OBJECTIVES

Solar Power System Objectives

Design Objectives

Objective 1: De-stress the local energy grid during peak use hours.

Power Electronics: Select a resilient programmable solar power system that is able to recognize off-peak, standard and peak usage hours and optimize energy use dynamically according to a set programmed instructions. The system should be able to incorporate a solar array, battery back-up and grid-tie components with antiislanding.

Solar Array: The solar array must maximize solar energy harvesting year round, though most especially during the time period between November and April as this is the period that the grid is most unstable. The solar panels should connect to the power electronics via a set of MPPT charge controllers to maximize solar input even in low light conditions. The solar array must be large enough to cover the majority, if not all of the morning peak energy usage requirements of the Client 2 facility.

Battery Bank: As there is a morning and afternoon peak demand usage period, a battery bank will be required in order to store solar energy during the morning solar peak hours and cover the afternoon peak energy period. The battery bank needs to be just large enough to cover the afternoon peak

usage period demand.

Objective 2: Design a back-up power solution should grid energy supply fail.

The Client 2 facility doesn't need to be able to run off-grid. However, it will be highly beneficial if key systems such as lighting, security E-fence for the animals and CCTV are able to continue running through the night should the grid fail and not recover quickly.

A separate emergency AC load panel can be installed to support such key systems. This panel will act as the standard AC load panel does in all cases except when grid power is cut, in which case the emergency AC load panel will be seen as the priority load by the central solar power system and draw its energy from the battery bank.

When grid energy is restored, facility loads connected to the emergency AC load panel will revert back to acting as a standard load powered by grid or solar input depending on the time of day and solar input levels. Contingency capacity should be considered when sizing the battery bank.



Objective 3: Create an income stream and reduce electricity expense for the Client 2 with the solar power system.

Energy cost savings and income stream can come from the following sources:

- Load Shifting (savings)
- Peak Shaving (savings)
- Feed In Tariffs (income)

Load Shifting: Once the energy usage load profile of the Client 2 facility is confirmed, it will need to be determined which loads can adjust usage periods in order to flatten the load profile and only maximize use of off-peak and standard hour rates when neither solar nor battery back-up are available. This will reduce or eliminate the need to pay peak usage hour electricity rates.

Peak Shaving: Reduce or eliminate peak energy usage during both peak periods by allowing solar input to support morning peak energy demand and excess solar and off-peak grid energy stored in the battery bank to support afternoon peak energy demand, shaving the requirement for grid support during peak demand hours.

Feed In Tariffs: Excess solar energy harvested above what is needed to support loads or charge batteries can be sold back to the grid.

Objective 4: Design a solar power system that is an impressive public relations piece.

Instead of closing off the solar power electronics in an electrical utility room, the system can be on display for visitors to the Client 2 to view as a showcase of the organization's commitment to sustainable animal care, their community and the environment.

Power electronics can be arranged in the back of the main building lobby along with a large 60" TV display showing the live operation of the solar power system in real time. The area around the electronics should be properly marked to observe proper safety protocols and be informative to the viewer at the same time.

Visitors will see the clean solar array on the main building roof, then walk inside the main door and see a state of the art solar power system displayed on the northern and western walls.

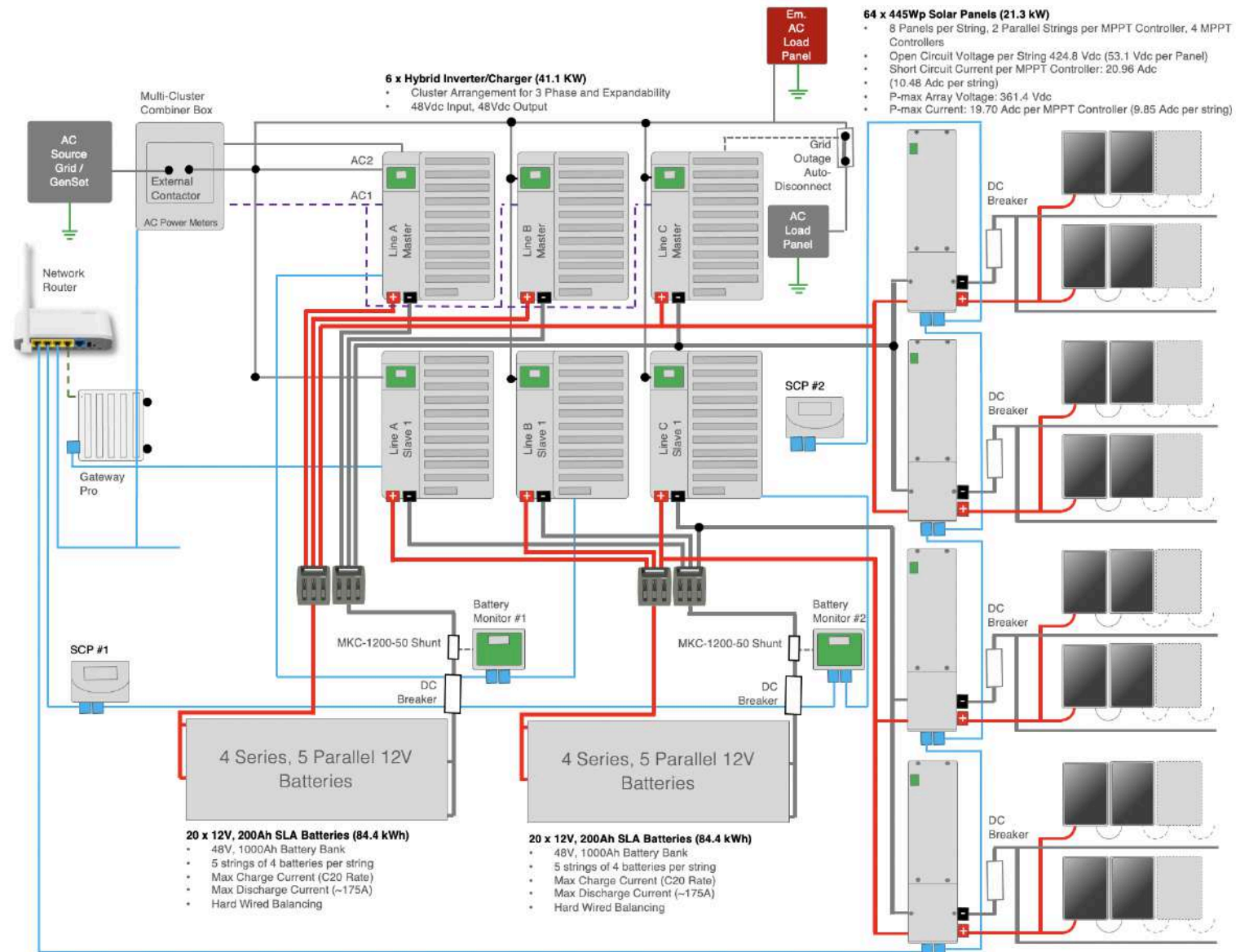


SYSTEM DESIGN

System Diagram

System Key Components

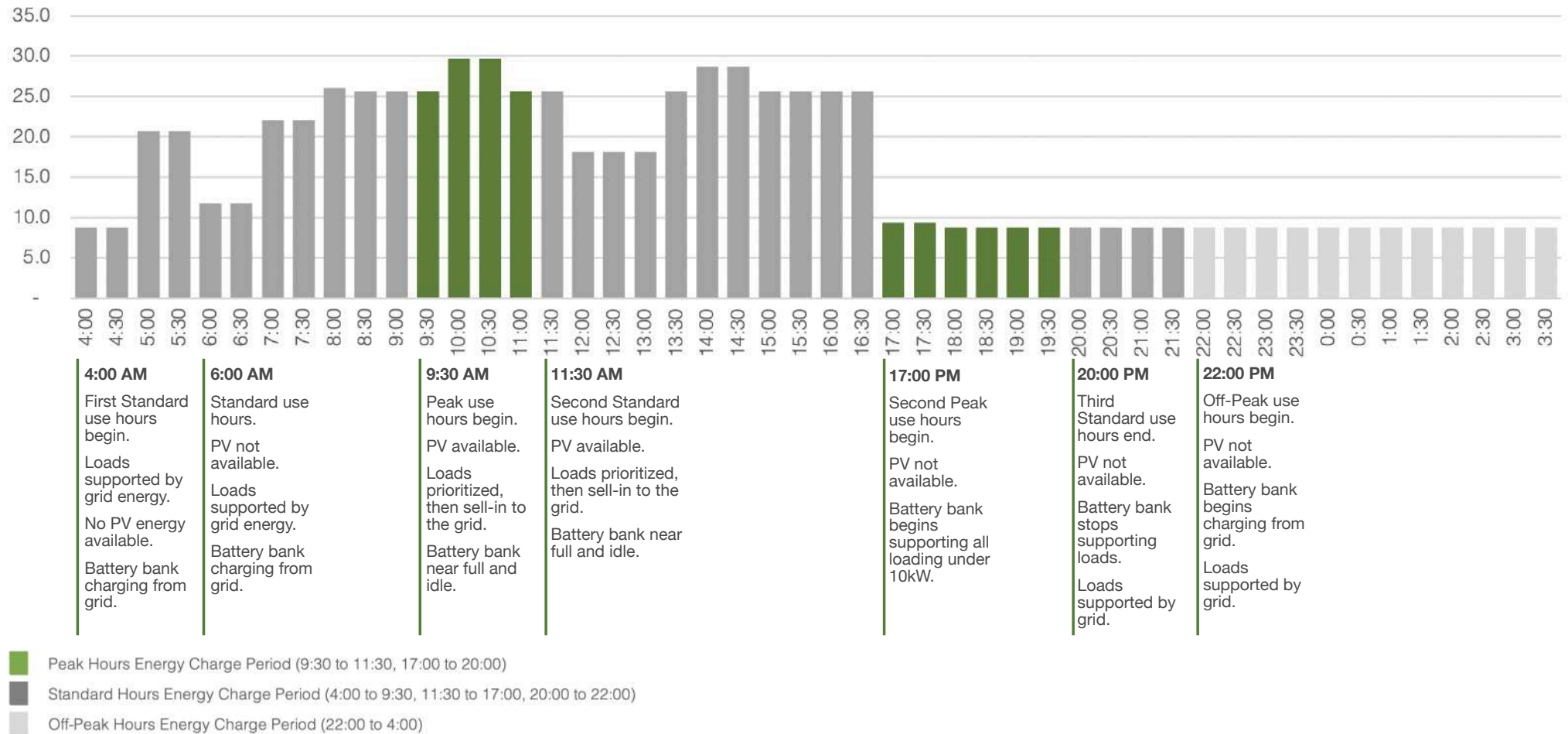
1. 6 X Schneider CONEXT™ XW+ 8548 E 230V Hybrid Inverter/Charger
Part Number: 865-8548-61
2. 4 X Schneider CONEXT™ MPPT 80 600 Solar Charge Controller
Part Number: 865-1032
3. 2 X Schneider CONEXT™ System Control Panel
Part Number: 865-1050-01
4. 2 X Schneider CONEXT™ Battery Monitor
Part Number: 865-1080-01
5. 1 X Schneider CONEXT™ Gateway Pro
Part Number: 865-1080-01
6. 2 X PowerLogic Panel Mount Meter (from Schneider)
Part Number: PM3250
7. Schneider CONEXT™ XW+ Power Distribution Panel - PDP
 - a. 1 x XW Power Distribution Panel:
Part Number: 865-1015
 - b. 6 x XW Connection Kit for INV2:
Part Number: 865-1020
 - c. 1 x XW Conduit Box:
Part Number: 865-1025
8. 11. 2 X Deltec Shunt
Part Number: MKC-1200-50
9. 4 x High Voltage DC Breakers (for PV Strings)
10. 13. 1 x Medium Voltage DC Breaker (for Battery Bank)
11. 64 x JAM78S10-445MRWp Solar Panels
12. 40 x 6FM200D-X Deep Cycle Batteries



FUNCTIONAL DESIGN

Load Support Profile

Energy Optimization Schedule

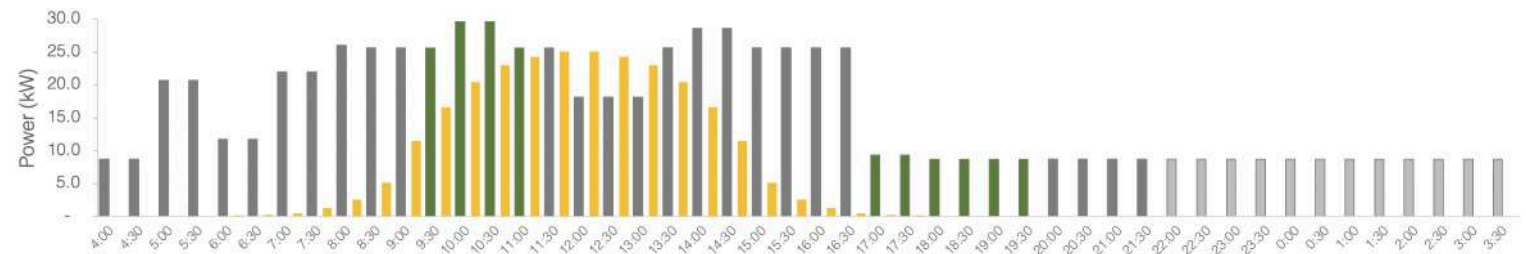


FUNCTIONAL DESIGN

Subsystem Load Support Profiles

Daily Load & Solar Profile

- Peak Hours Energy Use
- Standard Hours Energy Use
- Off-Peak Hours Energy Use
- Solar Energy Input



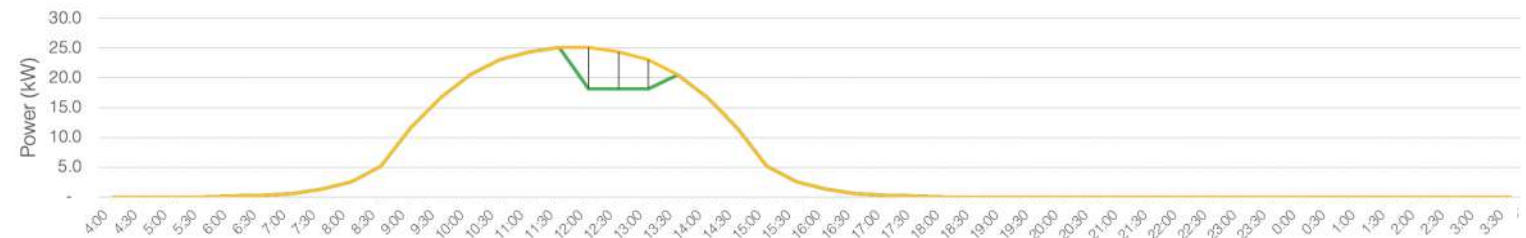
Daily Battery Energy Profile

- Battery Bank Charge from Grid
- Load Support by Battery Bank



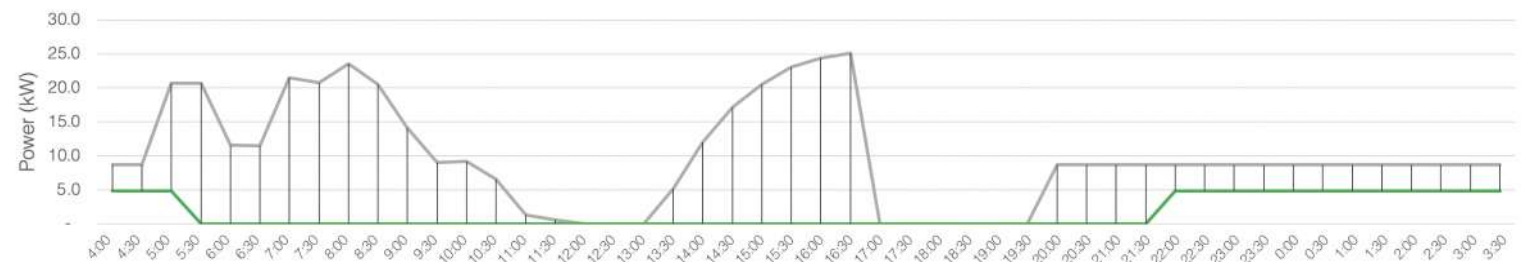
Daily Solar Energy Profile

- Solar Energy Profile
- Solar Energy Sold to Grid



Daily Grid Energy Profile

- Grid Load Support Profile
- Battery Bank Charge from Grid



INSTALLATION

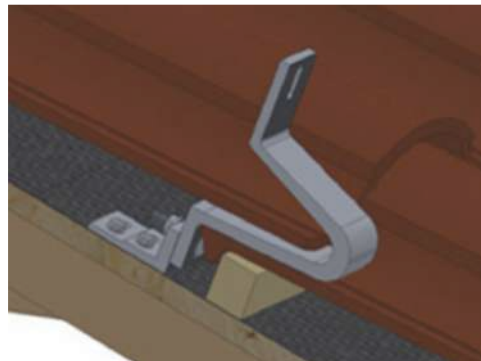
Solar Array



The 64 pieces of JA 78S10 445Wp solar panels are installed in an even array on the roof of the main Client 2 building. Shadows cast by roof features have been considered in the placement of the solar panels to avoid low harvesting levels as much as possible when the sun is low. A ladder was installed on the back of the building for quick roof access. The panels were mounted on the roof by a type of bracket that does not require cutting of roof tiles.

Orientation of the panels is due south at an approximate 54° angle (rather than 78°) which will allow for ideal solar harvesting in the winter months (November through April) rather than year round.

Solar array output at STP is 28.48kW and 22.78kW nominal (typical) in full sun. The array is made up of four subarrays comprising of two strings of eight solar panels each. There is space left on the south facing roof for approximately 30 more of the same solar panel increasing the current array output by 47% equating to 13kW. In this case, the number of solar charge controllers would need to scale up accordingly.



INSTALLATION

Solar Power Electronics

The six Schneider CONEXT™ XW+ inverters are arranged in clusters of three with one power distribution panel for each cluster. The four 80 600 MPPT charge controllers receiving energy from the solar array are situated under the stairwell on the northern (back) wall. The inverters are connected in two clusters, each cluster a set of three inverters. The two inverter clusters are mounted in a position that will allow for the system to be expanded by at least six additional inverters (two additional clusters) with room to spare for heat ventilation hardware on all four of the of the inverter clusters. This would double the power capacity of the system from the current 41.1kW to 82.2kW.

The four MPPT charge controllers are similarly arranged in that four additional charge controllers can be installed to the right of the current four charge controllers if needed.

Inverter clusters and charge controllers were installed on either side of the buildings main electrical panel, next to which was installed the necessary Gateway Conext and internet wireless routers.

To the right of the charge controllers a panel housing the power meters, battery monitors and system control panels was installed. The emergency AC load panel was not installed with the system. The display monitor was also installed to the left of the inverter clusters.



INSTALLATION

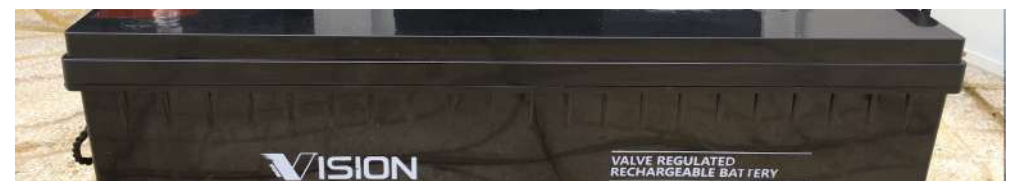
Solar Battery Bank

The battery cabinet is a standard outdoor electrical cabinet and includes a total of four shelves plus the floor of the enclosure. The 12V 200Ah deep cycle SLA batteries were placed in two rows of four per shelf totaling forty batteries and 96kWh of energy capacity. At 50% DOD, available energy storage capacity for use is 48kWh. Connected in two banks of 48V, each bank supports each of the two inverter clusters respectively.

Battery connections were organized in such a way as to allow the batteries to equalize as they charge without the need of this function from the charger inverters. Additionally, the primary positive and negative terminals are connected in adjacent orientation from one another in order to promote uniform charging and discharging.

Given the high winds, heavy rains and high heat the region receives throughout the year, additional measures were taken to protect the physical batteries. While ventilation is part of the enclosure, an additional larger vent was installed at the top of the cabinet. Furthermore, a covering structure was erected over the battery cabinet to protect from torrential rains and direct heat from the sun.

The original design called for refurbished lead acid wet batteries to be used as a means of supporting the environment further, though no reliable source was identified and deep cycle SLA batteries were specified instead.



FINANCIAL ANALYSIS

Impact on Loading, Expenses & Cash Flow

Note: All values in US dollars unless otherwise stated.

Grid Loading, Energy & Expense Reductions

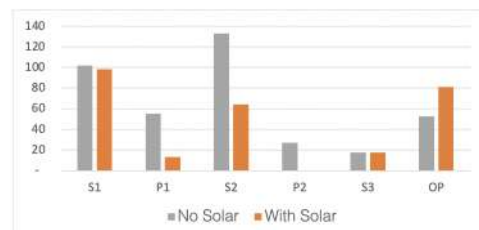
Before Solar Installation

Unit	Rate Periods						Daily Subtotal
	S1	P1	S2	P2	S3	OP	
kWh/day	101.96	55.34	132.92	26.84	17.47	52.41	387
USD/day	\$ 11.09	\$ 10.35	\$ 14.45	\$ 5.02	\$ 1.90	\$ 3.47	\$ 46.28

With Solar Installation

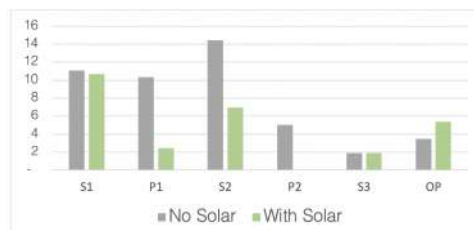
Unit	Rate Periods						Daily Subtotal
	S1	P1	S2	P2	S3	OP	
kWh/day	98.45	13.04	64.02	-	17.47	81.21	274
USD/day	\$ 10.70	\$ 2.44	\$ 6.96	\$ -	\$ 1.90	\$ 5.37	\$ 27.38

Energy Usage (kWh/day by Period)



Total Reduction: 386 kWh - 274 kWh = **113k Wh/day**

Energy Expense (USD/day by Period)



Total Reduction: US\$ 46.3 - US\$ 27.4 = **US\$ 18.9/day**

Local Energy Rates

	Wholesale charge for other purposes	Rate		Period		ID
		VND/kWh	USD/kWh	Start	End	
1	Standard hours 1	\$ 2,528	\$ 0.109	4:00	9:00	S1
2	Peak hours 1	\$ 4,349	\$ 0.187	9:30	11:00	P1
3	Standard hours 2	\$ 2,528	\$ 0.109	11:30	16:30	S2
4	Peak hours 2	\$ 4,349	\$ 0.187	17:00	19:30	P2
5	Standard hours 3	\$ 2,528	\$ 0.109	20:00	21:30	S3
6	Off-peak hours	\$ 1,538	\$ 0.066	22:00	3:30	OP
7	Feed-in-Tariff for Region: Qualifies for "Solar power project with an integrated storage system"	\$ 1,909	\$ 0.082			

Source: <https://en.evn.com.vn/d6/news/WHOLESALE-ELECTRICITY-TARIFF-9-28-260.aspx>

Source: <https://www.xe.com/currencyconverter/convert/?From=VND&To=USD>

Savings, Income, Depreciation & Cash Flow

Savings & Income from Solar Power

	S1	P1	S2	P2	S3	OP	Daily Subtotal
Solar							
Output 1 (USD/day) - Loads	\$ 1.16	\$ 7.91	\$ 7.49	\$ 0.04	\$ -	\$ -	\$ 16.61
Input 1 (USD/day) - Battery Bank	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Input 2 (USD/day) - Grid-Sell-In	\$ -	\$ -	\$ 0.74	\$ -	\$ -	\$ -	\$ 0.74
Subtotals:	\$ 1.16	\$ 7.91	\$ 8.23	\$ 0.04	\$ -	\$ -	\$ 17.35
Battery Bank							
Input 1 (USD/day) - Solar	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Input 2 (USD/day) - Grid	\$ 0.78	\$ -	\$ -	\$ -	\$ -	\$ 1.91	\$ 2.69
Output (USD/day) - Loads	\$ -	\$ -	\$ -	\$ 5.02	\$ -	\$ -	\$ 5.02
Subtotals:	\$ 0.78	\$ -	\$ -	\$ 5.02	\$ -	\$ 1.91	\$ 2.33
Grid							
Output 1 (USD/day) - Loads	\$ 9.92	\$ 2.44	\$ 6.96	\$ -	\$ 1.90	\$ 3.47	\$ 24.69
Output 2 (USD/day) - Battery Bank	\$ 0.78	\$ -	\$ -	\$ -	\$ -	\$ 1.91	\$ 2.69
Subtotals:	\$ 10.70	\$ 2.44	\$ 6.96	\$ -	\$ 1.90	\$ 5.37	\$ 27.38

- Savings from Solar Harvesting: US\$16.61 daily average
- Savings from Peak Shaving: US\$ 2.33 daily average
- Income from Feed-In-Tariff: US\$ 0.74 daily average

Depreciation Calculation

Component	Cost (USD)	Life (years)	Salvage Value (USD)	Annual Dep. Cost (USD)	Annual Subtotal
Power Electronics	\$ 26,637	10	\$ 1,000	\$ 2,564	
Batteries	\$ 12,522	10	\$ 1,248	\$ 1,127	\$ 4,043
Solar Panels	\$ 8,792	25	\$ -	\$ 352	

Annual Cash Flow (2021): Average Annual Expense Reduction **US\$ 3,020**

	Jan-21	Feb-21	Mar-21	Apr-21	May-21	Jun-21	Jul-21	Aug-21	Sep-21	Oct-21	Nov-21	Dec-21
Cash Inflows												
Donation	\$ 61,177	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Electricity Savings	\$ 576	\$ 576	\$ 576	\$ 576	\$ 576	\$ 576	\$ 576	\$ 576	\$ 576	\$ 576	\$ 576	\$ 576
Energy Sold	\$ 23	\$ 23	\$ 23	\$ 23	\$ 23	\$ 23	\$ 23	\$ 23	\$ 23	\$ 23	\$ 23	\$ 23
Subtotal:	\$ 61,775	\$ 599	\$ 599	\$ 599	\$ 599	\$ 599	\$ 599	\$ 599	\$ 599	\$ 599	\$ 599	\$ 599
Cash Outflows												
System Cost	\$ 61,177	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Depreciation	\$ 337	\$ 337	\$ 337	\$ 337	\$ 337	\$ 337	\$ 337	\$ 337	\$ 337	\$ 337	\$ 337	\$ 337
Operation	\$ 10	\$ 10	\$ 10	\$ 10	\$ 10	\$ 10	\$ 10	\$ 10	\$ 10	\$ 10	\$ 10	\$ 10
Subtotal:	\$ 61,523	\$ 347	\$ 347	\$ 347	\$ 347	\$ 347	\$ 347	\$ 347	\$ 347	\$ 347	\$ 347	\$ 347
Total Cash Flow												
Cash Flow	\$ 252	\$ 252	\$ 252	\$ 252	\$ 252	\$ 252	\$ 252	\$ 252	\$ 252	\$ 252	\$ 252	\$ 252
Balance	\$ 252	\$ 503	\$ 755	\$ 1,007	\$ 1,258	\$ 1,510	\$ 1,762	\$ 2,013	\$ 2,265	\$ 2,517	\$ 2,768	\$ 3,020

PROJECT CONCLUSION

Review & Summary

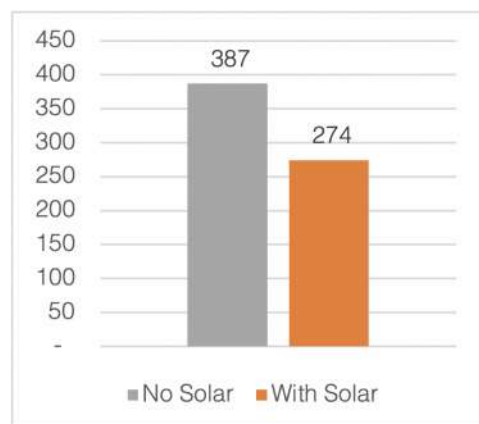
Objectives Met

Of the four objectives outlined to meet the needs of the expanding Client 2, three were met, while the fourth can be implemented at any time should the Client 2 team require.

Objective 1: De-stress the local energy grid during peak hours.

The solar power system reduced total energy demand from the grid by 29.2% on average. This is a reduction of 113kWh daily and 41.25kWh annually on average. Additionally, in times of low load demand, excess energy will be fed back to the energy grid further supporting external energy demand during the first of the two daily peak hours.

Average Energy Usage from Grid (kWh/day)



Objective 2: Design a back-up power solution should grid energy supply fail.

While the capability is present with the installed solar power system to meet this objective, the Client 2 team decided not to connect emergency loads to a separate panel to facilitate an emergency back up system. The 32.25kWhrs of available energy stored in the battery bank can allow for 64 hours (almost 2 days) of E-fence operation should all other power supplies fail. CCTV would be the next critical system to be connected to the emergency load panel, then emergency lighting.

Objective 3: Create an income stream and reduce electricity expense for the Client 2 with the solar power system.

Both aspects of this objective with the system installation as designed are met. Solar harvesting and peak shaving combined now support US\$6,913/year of energy demand savings and US\$270/year are fed back to the grid.

Depreciation of the solar array, solar power electronics and batteries equate to US\$4,043 annually. Total annual cash flow after depreciation and US\$120/year system operational cost is US\$3,020.

Objective 4: Design a solar power system that is an impressive public relations piece.

The solar power system selected is manufactured by Schneider Electric, a global brand well recognized for high quality and is traded publicly. The system is arranged between the western and northern walls of the Client 2's main building lobby and set up as a display piece for visitors. The system is connected to a large TV display showcasing the live performance of the solar power system via the Schneider Conext Gateway control and Insight 2 monitoring systems.



System Expandability

The system is designed to allow for ready expansion to near double its current size. Solar panel array can expand from its current 64 panels adding maximum 38 panels of the same specification with the remaining space available on the south facing facility roof. This would also require 3 to 4 additional MPPT 80 600 Schneider

Electric solar charge controllers. This expansion would cost approximately US\$5,125, US\$5,220 and US\$9,228 for the charge controllers, solar panels and a third inverter cluster respectively. This expansion would increase average annual cash flow by approximately US\$2,000, making a total of US\$5,020. More energy storage can be purchased and connected to the current system with little additional hardware other than wiring, breakers and proper enclosures.

Operation & Monitoring

Operation of the system is handled by computer interface by the Schneider Conext Gateway system and monitoring can be done by the Conext Insight 2 web portal, also provided by Schneider Electric Solar.

Wrap Up

The 21kW solar power array and 41kW grid tied solar inverter clusters with battery backup installed at the Client 2 facility is state of the art and will allow the organization to take advantage of the abundant solar energy available in Vietnam. The system meets Client 2's requirements and more. As long as it is well maintained, the system should continue to be a high performing unit that will continue to save energy and impress visitors to the Client 2 for years to come. The system was commissioned and put into operation on December 23rd, 2020.



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